

AARON FOOTE

 [aaronfoote31@gmail.com](#) 

EDUCATION

Wesleyan University, Middletown, CT

September 2024 - May 2025

M.A. in Computer Science

GPA: 4.22/4.00

Thesis: *Diagnosing Biased Feature Importance Scores in Tree-Based Models*

Wesleyan University, Middletown, CT

September 2020 - May 2024

B.A. with High Honors in Computer Science, Minor in Data Analysis

GPA: 3.87/4.00

Thesis: *On the Complexity and Threshold Behavior of Playing Nonogram Puzzles*

Relevant Coursework:

Mathematical Statistics, Survival Analysis, Statistical Learning, Algorithms, Applied Bayesian Analysis, Artificial Intelligence, Applications of Linear Algebra in Machine Learning, Real Analysis, Probability Theory, Data Structures

RESEARCH EXPERIENCE

Graduate Research Assistant – Interpretable ML

June 2024 - Present

Wesleyan University

Middletown, CT

- Developed a hypothesis test under minimal assumptions to evaluate feature importance methods for bias
- Validated the method through extensive simulation and application to real datasets in Python and R
- Integrating dimension reduction techniques to extend the test to higher dimension
- Investigating applications to gene expression data to produce reliable, efficient feature importance scores to improve the interpretability of black-box ML models for their application in medicine

Research Assistant – Algorithms, Computational Complexity

September 2022 - Present

Wesleyan University

Middletown, CT

- Established complexity results and phase transition behavior for the solving of combinatorial puzzles
- Developed solvers in C/bash for high-performance parallel computing cluster
- Studied the complexity of puzzles published online, scraping thousands of published puzzles with Python
- Modeling human solving strategies to understand the connection between human and computational difficulty
- Formalizing and proving complexity results for problems related to generating puzzles

Research Assistant – Machine Learning in Sports

January 2024 - June 2024

Wesleyan University and Loyola University Chicago

Middletown, CT and Chicago, IL

- Produced an intuitive evaluation tool for NFL defenders using spatiotemporal player tracking data
- Estimated play outcomes using conditional density estimation random forest method, then scored defenders on minimizing expected points, assigning credit by weighted distance
- Designed and deployed an NCAA March Madness prediction model using Cox proportional hazards, optimizing performance through bootstrapping coefficients
- Explored time-varying coefficients Cox model and random versus deterministic bracket completion

Research Fellow – Reinforcement Learning

May 2023 - December 2023

Wesleyan University

Middletown, CT

- Formalized poker strategies, implemented evolutionary game theory algorithms, and simulated matches with reinforcement learning agents to identify effective strategies

Summer Research Fellow – Computer Vision, Natural Language Processing

June 2021 - September 2021

Wesleyan University

Middletown, CT

- Developed a pipeline to extract key frames from political ads using corner detection algorithms and OCR, cutting the financial cost of analyzing political ad data by a factor of 100
- Implemented an algorithm to clean output from OCR into time-ordered, repeat-free output using string clustering
- Attended statistical methods workshops covering log-linear modeling (Poisson, negative-binomial, ordinal), longitudinal analysis, regression post-estimation, survival analysis, and exploratory data analysis

TEACHING EXPERIENCE

Head Quantitative Analysis Tutor

Wesleyan University

April 2023 - May 2025

Middletown, CT

- Hold 10+ hours of weekly office hours open to the entire university, assisting with coursework, projects and research
- Provide one-on-one statistical consulting for undergraduate and graduate thesis projects, ensuring methodological rigor
- Guide students in managing and analyzing data, employing and interpreting statistical techniques, and visualizing their work with R, STATA, Python, SPSS, SAS, and Excel
- Topics tutored: data visualization, survival analysis, machine learning, hypothesis testing, regression, regression diagnostics, longitudinal data analysis, econometrics, data engineering
- Develop a schedule each semester, lead weekly meetings, and consult with professors

Statistics/Data Analysis Teaching Assistant

Wesleyan University

January 2024 - May 2024

Middletown, CT

- **Survival Analysis** (Spring 2024): Clarified concepts in class, guided students on lab work, and consulted with students on final projects involving topics such as censoring, hazard and survival functions, Kaplan Meier estimates, Cox proportional hazards model, time-varying covariates, parametric survival analysis, and accelerated failure-time distributions
- **Applied Data Analysis** (Spring 2024): Mentored six students for five hours per week at every stage of a research project (identifying research questions, preparing and visualizing data, running statistical tests, and linear/logistic regression)

Computer Science Teaching Assistant

Wesleyan University

September 2022 - December 2024

Middletown, CT

- **Algorithms** (Fall 2024): Instructed a dozen students once a week on material including analysis of algorithms, divide-and-conquer, dynamic programming, greedy algorithms, branch and bound, and approximation algorithms
- **Theory of Computation** (Fall 2022): Instructed students once per week on material including finite automata, Turing machines, NP-completeness, polynomial time reductions, space complexity, and approximation algorithms

RESEARCH PRESENTATIONS

Foote, A. and Krizanc, D. (2025) "TRIP: A Nonparametric Test to Diagnose Biased Feature Importance Scores." *IJCAI Workshop on Explainable Artificial Intelligence* (Under Review)

Foote, A. and Matthews, G. (2024). "Continuous Within-Play Estimation of Run-Stopping Contribution in American Football." *Cascadia Symposium on Statistics in Sports* (September 2024, Conference Poster).

Foote, A., Gooyabadi, M. and Addleman, N. (2023). "Factors in Learning Dynamics Influencing Relative Strengths of Strategies in Poker Simulation." *Games*. doi:10.3390/g14060073.

Foote, A. and Oleinikov, P. (2021) "Efficiently Extracting Text From Political Ad Videos." *CIS Summer Research Symposium* (July 2021, Poster).

AWARDS

Maureen Snow Prize (July 2023): Runner-up among 200 students, given for most effective communication of research to a broad audience

Baker '64 Collabria Fellowship in Data Analysis (April 2023) – \$5,000

Best Data Story Award (April 2023): Awarded at Northeast Division ASA DataFest Competition

QAC Apprentice Grant (April 2021) – \$5,400